AP Precalculus — Unit 1, Lesson 1.10

Guided Notes — Answer Key

Name: ____________________________ Date: ________________ Period: ________

Primary Objective

By the end of this lesson, I will be able to...

1. Determine whether a rational function has a **hole** or a **vertical asymptote** at a shared zero by comparing **multiplicities**. *(LO 1.10.A)*
2. Find the **coordinates** (c, L) of a hole and express the result using **limit notation**. *(LO 1.10.A)*

Vocabulary & Key Concepts

hole in a graph:

A missing point (c, L) where $r(c)$ is undefined but $\lim_{x\to c} r(x) = L$ exists.

removable discontinuity:

Another name for a hole; can be "removed" by redefining the function at one point.

common factor:

A polynomial factor shared by numerator and denominator; canceling it reveals a hole.

limit of a function at a point:

$\lim_{x\to c} r(x) = L$**: outputs of r approach L as inputs approach c.**

Hook

Imagine your GPS loses signal for exactly one moment — at $t = 5$ seconds — but your position function is defined everywhere else. The output *at* $t = 5$ is missing, but by looking at outputs for inputs very close to 5, we can determine where the position function "should" be. This missing-but-predictable point is called a **hole** in the graph of a rational function.

Holes vs. Vertical Asymptotes

Let $r(x)$ be a rational function with a shared zero at $x = a$. Let m = multiplicity of a in the **numerator** and n = multiplicity of a in the **denominator**.

Condition	Feature at $x = a$	Example
$m \geq n$	**Hole**	$\frac{(x-2)^2}{x-2}$: hole at $x = 2$
$m < n$	**Vertical Asymptote**	$\frac{x-2}{(x-2)^2}$: VA at $x = 2$
a only in denominator	**Vertical Asymptote**	$\frac{5}{x-2}$: VA at $x = 2$

Key idea: A hole occurs when the numerator and denominator *share* a zero *and* the numerator's multiplicity is **greater than** or **equal to** the denominator's.

Finding the Coordinates of a Hole

4-Step Procedure:

1. **Factor** the numerator and denominator.
2. Identify **shared** zeros and compare multiplicities.
3. **Cancel** the common factor from numerator and denominator.
4. Substitute $x = c$ into the **simplified** form to find L.

Example 1. $r(x) = \dfrac{x^2 - 4}{x - 2}$

Step 1 — Factor: $r(x) = \dfrac{(x-2)(x+2)}{x-2}$

Step 2 — Shared zero: $x = 2$ Num. mult: **1** Denom. mult: **1** Since $m = n$, this is a **hole**.

Step 3 — Cancel: simplified form $= x + 2$

Step 4 — Substitute $x = 2$: $L = 2 + 2 = 4$

Hole at: $\left(\mathbf{2}, \mathbf{4}\right)$

Limit Notation for Holes

If a rational function r has a hole at (c, L), we write:

$$\lim_{x \to c} r(x) = L$$

This means: as the input x gets **arbitrarily close** to c (from either side), the output $r(x)$ gets **arbitrarily close** to L.

Both one-sided limits agree: $\lim_{x \to c^-} r(x) = L$ and $\lim_{x \to c^+} r(x) = L$

Example 2. $r(x) = \dfrac{x^2 - 1}{(x-1)(x+2)}$

Shared zero at $x = 1$. Simplified form: $\dfrac{x+1}{x+2}$

$L = \dfrac{2}{3} = \dfrac{2}{3}$

Limit statement: $\lim_{x \to 1} r(x) = \dfrac{2}{3}$ Hole at: $\left(\mathbf{1}, \dfrac{2}{3}\right)$

Is there a VA? **Yes** If so, where? $x = -2$